FIR VisionAI – Complete Architecture & Step-by-Step User Guide

Unified FIR & AI-Powered Case Intelligence Platform | Andhra Pradesh Prohibition & Excise Department

1. System Architecture Overview

Users (Frontend)

■ Excise Officer

- ■ Police Officer
- ■ Investigation Team
- ■ Admin / Supervisors
- ■ Legal & Policy Team
- ■ Department Headquarters

HTTPS →

Frontend (React)

■ Web Application (React 18)

- • Interactive Dashboard
- • Case Management Workspace
- • New FIR Intake & Evidence
- • AI & Legal Intelligence
- • Knowledge Base Hub
- • AI FIR Chatbot (RAG)
- • Investigation Graph Tools
- • Administration & Roles
- • Reports & Downloads

Backend (FastAPI)

■ FastAPI (Python 3.12)

- • Auth & RBAC Authorization
- • FIR & Case Management APIs
- • Document Upload & OCR APIs
- • AI Extraction & Validation
- • Legal Intelligence APIs
- • RAG / Chatbot APIs
- • Entity Matching & Graph APIs
- • Administration & Audit APIs
- • Multi-format Export APIs

Async →

Processing Services

■ Async Processing Workers

- • Image/PDF Pre-processing
- • Multilingual OCR (TE/HI/EN)
- • Layout & Zone Segmentation
- • Field Validation Engine
- • Legal Section Matcher
- • Entity Recognition (NER)
- • RAG Indexing & Chunking
- • Background Job Queues

Data Layer (Supabase)

■ Supabase (PostgreSQL)

- • Application Relational DB
- • pgvector Embedding Store
- • Evidence Storage Buckets
- • Row Level Security (RLS)
- • Realtime Event Bus
- • Automatic Daily Backups
- • Complete Audit Trail Tables

AI & External Services

■ Foundation AI & Models

- • Gemini 1.5 Pro / Flash Vision
- • Tesseract Indic OCR Engine
- • Text Embeddings (1536-dim)
- • Official Legal Acts Corpus
- • Nominatim OpenStreetMap
- • AP 26-District Centroid DB
- • Google Maps Directions API

2. End-to-End Workflow (6 Main Stages)

1. Evidence Upload & OCR
- • Upload FIR (PDF / Image)
- • Pre-process & Despeckle
- • Extract Multilingual Text
- • Store Raw Evidence PDF
- • Bounding-box Inspection

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2. Officer Review & Sign-Off
- • Verify Extracted Fields
- • Correct Uncertain Data
- • Check Original FIR Page
- • Mandatory Human Sign-off
- • Approve or Reject Draft

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3. AI & Legal Intelligence
- • Suggest Legal Sections
- • Match BNS / AP Excise
- • Confidence Scoring
- • Statutory Reference Links
- • Essential Ingredient Checks

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4. Investigation Tools
- • Person / Vehicle Entities
- • Pin Exact Incident Map
- • Criminal Network Graph
- • Cross-case Connections
- • Investigation Timeline

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5. Administration & FIRs
- • Jurisdiction & Role Scopes
- • Manage Knowledge Base
- • Search Historic FIRs
- • Immutable Audit Trail
- • System Health Monitor

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6. Repeat Offender Intelligence
- • Cross-Station Repeat Match
- • Link Alias & Vehicles
- • Network Visualization
- • Prior Conviction Insights
- • Seizure Pattern Analysis

3. Frontend - Key Pages and Buttons

Dashboard	Total cases, under review, approved, pending drafts. Filtered
New Case Intake	Upload handwritten/printed FIR scans, monitor OCR progress.
Case Management	Search, filter by station/year/status, open evidence workspace
AI & Legal Intel	View extracted sections, statutory penalties, legal citations.
Knowledge Base	8 authorized categories: Official Laws, Circulars, SOPs, Seizu
AI FIR Chatbot	Ask case questions, legal provisions, cross-case queries with
Investigation Tools	Entity graph, repeat offender detection, incident location map
Administration	Officer roles, jurisdiction scopes (State/Dist/Station), audit
Downloads & Reports	Export Case Diary, Chargesheet in DOCX/PDF, architecture guide

4. Database Schema (Supabase - Key Tables)

users
id (PK)
username
role (State/Dist/Stn)
jurisdiction
full_name
status

cases / firs
id (PK)
fir_number
station
district
status
incident_location
created_at

fir_documents
id (PK)
case_id (FK)
file_name
storage_path
ocr_text
page_count

fir_extractions
id (PK)
case_id (FK)
field_name
field_value
confidence
verified

knowledge_base
id (PK)
title
category (1-8)
file_path
content_text
created_at

kb_chunks
id (PK)
doc_id (FK)
chunk_text
embedding (1536)
metadata_json

chat_history
id (PK)
user_id (FK)
source_type
question
answer
citations

audit_logs
id (PK)
actor_id (FK)
action_type
object_type
object_id
timestamp

5. AI & RAG Pipeline

1. Multi-Category Ingestion
Upload official laws, circulars, SOPs, seizures, FIRs

2. Cleaning & Chunking
Normalized text extraction, chunking with overlap

3. Embedding Generation
Gemini / text-embedding-3 vectors (1536-dim)

4. Indexing in Supabase
Stored in pgvector with HNSW cosine distance index

5. Hybrid Vector Retrieval
Balanced cosine retrieval across 8 authorized scopes

6. Context Grounding
Strict prompt template enforcing evidence boundaries

7. Citations & Verification
Answers returned with exact document name & citations

6. Security & Control

■ Supabase Authentication
JWT Bearer tokens with auto-refresh

■ Role-Based Access (RBAC)
STATE, DISTRICT, and STATION tiers

■ Jurisdiction Scoping
Officers see only authorized cases

■ Row Level Security (RLS)
Postgres kernel-enforced data rules

■ Signed Evidence URLs
Direct access tokens for evidence PDFs

■ Immutable Audit Trail
All edits, approvals & exports logged

■ End-to-End Encryption
TLS 1.3 in-transit and AES-256 at-rest

■ No Auto-Guilty Finding
AI assists only; human officer sign-off

7. Document Processing Pipeline (From Upload to Final Output)

1. Upload FIR (PDF / Image) → **2. Storage Bucket** & Checksum → **3. Deskew, Denoise** & Binarize → **4. Multilingual OCR** (TE / HI / EN) → **5. Field Validation** Confidence Score → **6. Officer Review** & Edit / Sign-Off → **7. Index for Search** (Keywords+Vectors) → **8. Intelligence Tools** & Generation

8. Final Outputs

- ■ Structured FIR JSON Record
- ■ Verified Legal Sections & Acts
- ■ AI Answers with Grounded Citations
- ■ Incident Location on Map & GPS
- ■ Case Diary & Chargesheet (DOCX/PDF)